The Role of Liquid Penetration in Detergency of Long-Chain Fatty Acid

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ABSTRACT: The detergency phenomena of a solid oily contaminant assessed with a quartz crystal microbalance (QCM) technique was discussed on the basis of surface energetics. Polyethylene, nylon 6, and cellulose acetate films were prepared on gold electrodes of the QCM by a spin-coating method. Arachidic acid was deposited onto the QCM with and without polymer films by the Langmuir–Blodgett (LB) technique. The QCM was then ultrasonically cleaned in aqueous solutions containing surfactant, ethanol, and alkali. The removal efficiency of the LB films of arachidic acid from the gold and polymer substrates was determined from the frequency change of the QCM due to cleaning. The efficiency was greatest for cellulose acetate, followed by nylon 6, gold, and polyethylene. For every substrate, the removal efficiency was found to increase with increasing surfactant or ethanol concentration. In the presence of sodium hydroxide, the removal efficiency further increased. The experimentally determined contact angles and surface free energies were influenced by surfactant and ethanol concentrations; however, they were not influenced by the addition of sodium hydroxide. This suggested the saponification of arachidic acid under alkali conditions was a major mechanism of fatty acid removal. The dependences of the kind of substrate and surfactant and ethanol concentrations on the removal efficiency were explained in terms of the free energy change resulting from penetration of the detergent solution between the arachidic acid film and the substrate in the zone of contact. Liquid penetration as well as saponification was found to be an oil removal mechanism in the present system.

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KEY WORDS: Contact angle, detergency, Langmuir–Blodgett films, liquid penetration, oily contaminant, quartz crystal microbalance, surface free energy.

The removal of contaminants from apparel and industrial products has commonly been carried out in liquid media such as water and organic solvents. Organic detergent solvents are regarded as toxic substances, environmental pollutants, and hazardous wastes (1,2); therefore the manufacture of trichlorotrifluoroethane (CFC113) and 1,1,1-trichloroethane have been discontinued. Conversely, water is harmless to living things and has no impact on the environment. However, it is difficult to remove oily contaminants with water because of their insolubility. Rolling up and emulsification are two major mechanisms for the removal of liquid oily contaminants in the cleaning process with aqueous surfactant solutions (3–5). For solid oily contaminants such as long-chain fatty acids, solubilization and liquid crystal formation are possible removal mechanisms above the critical micelle concentration (CMC). Below the CMC, another removal mechanism in common with particulate soil may be operative because solid oily contaminants are not transformable. Applying these mechanisms would assist the development of an advanced aqueous detergent technique.

Recently, the application of the quartz crystal microbalance (QCM) method (6) to the evaluation of detergency of oily contaminants has been proposed (7,8). The QCM can detect the mass deposited onto its gold electrodes from the frequency shift. Removal efficiency of oily contaminants from gold electrodes of the QCM in aqueous solutions was successfully determined from a change in frequency of the QCM owing to oil removal. The results suggested that the removal of a solid oily contaminant from the gold substrate was characterized largely by liquid penetration into the contaminant/substrate interface. To give sufficient evidence for the liquid penetration mechanism in the oil-cleaning process, the removal behavior from various substrates merits thorough investigation.

In this paper, polyethylene, nylon 6, or cellulose acetate films were prepared on the gold electrode of the QCM by a spin-coating technique. Two solid-condensed monolayers of arachidic acid on water were transferred onto the QCM with and without the spin-coated film to act as an oily contaminant (9). The removal efficiency of the Langmuir–Blodgett (LB) films of arachidic acid from each substrate was investigated in a model aqueous detergent solution containing surfactant, ethanol, and alkali. The effects of the kind of substrate, of sodium dodecyl sulfate (SDS) and ethanol concentrations, and of the addition of alkali on the removal efficiency were analyzed from the viewpoint of saponification and liquid penetration.

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Abbreviations: CMC, critical micelle concentration; LB, Langmuir–Blodgett; QCM, quartz crystal microbalance; SDS, sodium dodecyl sulfate.

EXPERIMENTAL PROCEDURES

Materials. Arachidic acid (Wako Chemical Co., Ltd., Tokyo, Japan) was used as a model oily contaminant. The AT-cut, 9 MHz QCM (USI Systems Co., Fukuoka, Japan) was used to prepare a model substrate. The gold electrodes of 4.5 mm diameter (~300 nm thick) were deposited on chrome underlayers (~15 nm thick) on both sides of the quartz crystal. The QCM was cleaned with water and acetone. The frequency of the QCM was monitored with a frequency counter (HP 53131A/132A; Hewlett Packard, Palo Alto, CA). The mass deposited on the gold electrode of the QCM was calculated from the frequency change of the QCM using the Sauerbrey equation (10,11). The surfactant used was SDS (Wako Chemical Co., Ltd.). The CMC of SDS in the presence of 1×10^{-3} mol/dm^3 sodium chloride was determined to be 7.6×10^{-3} mol/dm^3 at 25°C by surface tension measurements using a pendant drop technique. The CMC value did not change significantly by the addition of 1×10^{-3} mol/dm^3 sodium hydroxide. Sodium chloride, sodium hydroxide, and ethanol were extrapure grade reagents and were used without further purification. Water was deionized and distilled using an Easy Pure RF apparatus and a Glass Still apparatus (Barnstead, Wiesbaden, Germany).

Sample preparation. Polymer substrates were prepared on the gold electrodes of the QCM as follows: polyethylene (Scientific Polymer Products Inc., Ontario, NY), nylon 6 (Scientific Polymer Products Inc.) and cellulose acetate (70% acetylated; Kishida Chemicals, Osaka, Japan) were dissolved in xylene, phenol/benzene (1:3, vol/vol), and acetone, respectively. Immediately after dropping 0.01 mL of each polymer solution (10 g/dm^3) on the gold electrode, the QCM was spun at *ca.* 8000 rpm for 10 s. The procedure was repeated twice. The mean thickness of each polymer film formed on the gold surface was estimated to be between 30 and 80 nm from the frequency change of the QCM by the spin-coating (8). The mean roughness of the film was determined to be several nanometers by atomic force microscopy images (8). It was confirmed that the detachment of the polymer film from the QCM in the following LB deposition and cleaning processes was negligibly small (8).

Deposition of arachidic acid LB films. The deposition of arachidic acid monolayers onto the substrate was performed using a computer-controlled LB film deposition apparatus (FW-2; Lauda GmbH, Lauda-Koenigshofen, Germany). The poly(tetrafluoroethylene) trough ($200 \times 600 \times 10$ mm^3) was filled with water controlled within 20 ± 0.1°C. Small drops of arachidic acid/benzene solution (0.2 g/dm^3) were dispersed on the water surface using a 1-mL syringe. After allowing 15 min for evaporation of benzene, the monolayer was compressed continuously with a barrier speed of 8 mm/min. At a surface pressure of 30 mN/m (12), two solid-condensed monolayers of arachidic acid were transferred onto the substrate on the QCM by the vertical dipping method at 5 mm/min. The LB film was expected to have a Y-type structure because the water surface was automatically compressed during both immersion and withdrawal of the substrate to keep surface pressure constant. The mass of the arachidic acid films deposited onto the substrate was calculated from the change in frequency before and after the LB deposition. The QCM with the arachidic acid films was aged for 1–2 d prior to removal experiments.

Removal of LB films of arachidic acid. In the removal experiments, the QCM with the LB films of arachidic acid was perpendicularly immersed in an aqueous SDS or ethanol solution, and ultrasonic waves were applied for the mechanical action of removal. All detergent solutions contained 1×10^{-3} mol/dm^3 sodium chloride as a supporting electrolyte. The removal experiments were carried out in the absence (pH 6.1 ± 0.2) and the presence of 1×10^{-3} mol/dm^3 of sodium hydroxide (pH 11.2 ± 0.2).

Although the application of the QCM method is expected to make possible *in situ* measurements of the detergency (13,14), the QCM response in contact with a liquid depends on various factors such as an electrode microstructure, morphology of surface films, interfacial liquid properties, and the surface–liquid coupling at the interface (15). In the present system, the QCM electrode was covered with a polymer film that has several nanometers in root square surface roughness and tens of nanometers in peak-to-valley distance, and; therefore the liquid motion generated by the oscillating surface becomes much more complicated. In addition, the mass of the surfactants adsorbed onto the substrate surface may affect the experimental result. Therefore, the frequency of the QCM was measured after displacing the solution that was taken up by pure water and drying in air. The mass of the arachidic acid films removed from the substrate was calculated from the frequency change due to the cleaning.

Determination of contact angles and surface free energies. The contact angles and the surface free energies of the detergent solutions were measured by the sessile drop and the pendant drop methods, respectively, using a video contact angle system (VCA2500; AST products, Billerica, MA) (16). A 2–3 μL drop was placed on the gold electrode of the QCM with and without the polymer or arachidic acid film. The drop was viewed from the front simultaneously using a CCD camera attached to a microscope. The drop images were digitized and stored in a computer. The contact angle was measured by placing markers around the circumference of the drop. At the initial stage of measurement, the contact angle decreased linearly with time whereas the drop diameter increased slightly, i.e., the water front advanced. Therefore, the contact angle obtained by extrapolating the measured angles to time zero was assumed to be the advancing contact angle.

In the measurement of the surface free energy, the solution was dispensed from the microsyringe to form a pendant drop, which was made as big as possible without letting it fall off the tip of the needle. The surface free energy was determined by pendant drop image analysis through video-image digitization and numerical curve-fitting using the Laplace equation of capillarity (17).

All experiments were carried out in a room maintained at 25 ± 1°C.

RESULTS AND DISCUSSION

Deposition of LB films onto substrates. The mass of arachidic acid transferred onto the substrates is presented in Table 1 (8). The transfer ratios onto various substrates in Table 1 were calculated from the theoretical mass of two dry monolayers of arachidic acid on two gold electrodes of the QCM (166 ng) (7). The monolayers of arachidic acid were transferred completely onto the gold and polyethylene surfaces as Y-type LB films (9). The transfer ratios for nylon 6 and cellulose acetate substrates showed poor adhesion of the arachidic acid monolayer to the substrate (9). Previously (12), it was observed with a Brewster angle microscope that the LB technique made arachidic acid deposit on the substrate much more uniformly than other techniques: Little deposition and patchy deposition were observed after dipping the substrate into the arachidic acid/benzene solution and after placing the solution on the substrate, respectively. Therefore, the LB technique may be one of the best choices for the deposition of oily contaminant even though the transfer ratio is less than unity.

Removal of LB films from substrates. It was previously observed that the removal efficiency of the arachidic acid LB film from the substrate increased with time and began to show saturation at 10 min (8). Therefore, the removal efficiency at 10 min was used as an apparent equilibrium value in the present study.

The apparent equilibrium removal efficiency of the arachidic acid LB films from various substrates is presented in Figure 1 as a function of SDS concentration. In all cases, the addition of 1×10^{-3} mol/dm^3 sodium hydroxide increased the removal efficiency of arachidic acid. In both the presence and absence of sodium hydroxide, the removal efficiency was greatest for cellulose acetate, followed by nylon 6, gold, and polyethylene. This order is not in contradiction with the transfer ratios of arachidic acid monolayers (Table 1), corresponding to the adhesive strength of the arachidic acid films to the substrates. The results in Figure 1 suggest that fatty acid is difficult to remove from metal and hydrophobic polymeric materials such as polyethylene.

The effect of ethanol concentration on the removal efficiency is shown in Figure 2. The removal efficiency increased considerably with increasing ethanol concentration, indicating that the polarity of the liquid medium had a great influence on fatty acid removal. In comparing the results in Figure 2 with those in Figure 1, a similar tendency was found

TABLE 1
Mass of arachidic acid LB films Deposited onto the Substrates Calculated from Frequency Change of QCM Owing to LB Deposition and Transfer Ratio of Arachidic Acid Monolayers on Water[a]

Substrates	Deposition / ng	Transfer ratio
Gold	151	0.91
Cellulose acetate	70	0.42
Nylon 6	115	0.69
Polyethylene	158	0.95

[a]LB, Langmuir-Blodgett; QCM, quartz crystal microbalance.

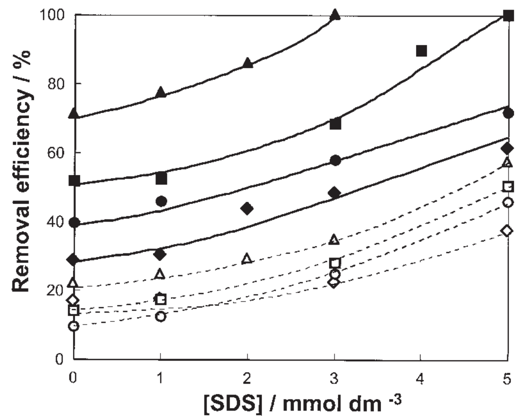

FIG. 1. Removal efficiency of arachidic acid films from gold (○, ●), polyethylene (◇, ◆), nylon 6 (□, ■), and cellulose acetate (△, ▲) substrates in the absence (open symbols) and presence (closed symbols) of 1×10^{-3} mol/dm^3 sodium hydroxide as a function of sodium dodecyl sulfate (SDS) concentration.

with respect to the concentrations of SDS and ethanol, the effect of the addition of sodium hydroxide, and other kinds of substrates.

Contact angles and surface free energies. Figures 3 and 4 present the advancing contact angle of the detergent solutions on the substrates and arachidic acid as functions of SDS and ethanol, respectively. In all cases, the advancing contact angle decreased linearly with increasing SDS or ethanol concentration. As shown in Figure 5, the surface free energies of the detergent solutions decreased with increasing SDS or ethanol concentration. The addition of 1×10^{-3} mol/dm^3 sodium hydroxide did not affect the advancing contact angle and the surface free energy of the detergent solution, although the results are not shown for simplicity.

The Lifshitz–van der Waals component and the Lewis acid and base parameters of the surface free energy of the sub-

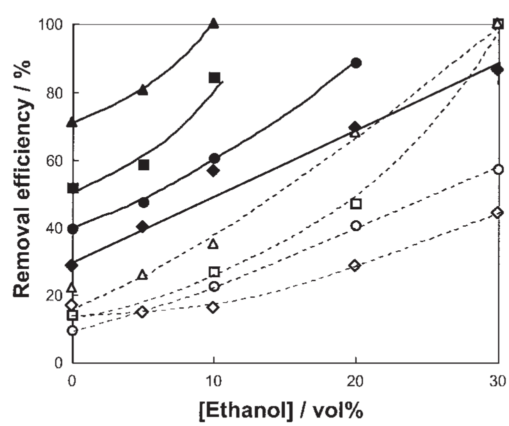

FIG. 2. Removal efficiency of arachidic acid films from gold (○, ●), polyethylene (◇, ◆), nylon 6 (□, ■), and cellulose acetate (△, ▲) substrates in the absence (open symbols) and presence (closed symbols) of 1×10^{-3} mol/dm^3 sodium hydroxide as a function of ethanol concentration.

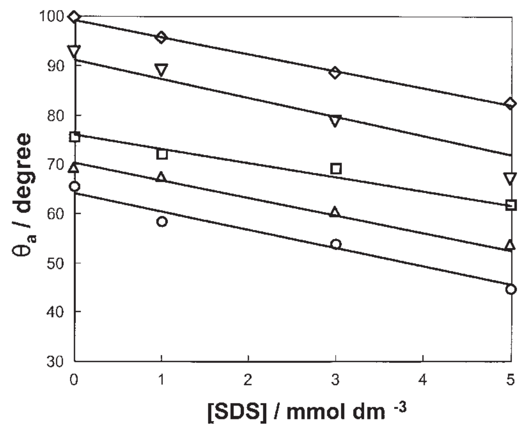

FIG. 3. Advancing contact angles (θ_a) of aqueous detergent solutions on gold (○), polyethylene (◇), nylon 6 (□), cellulose acetate (△), and arachidic acid (▽) as a function of SDS concentration. For abbreviation see Figure 1.

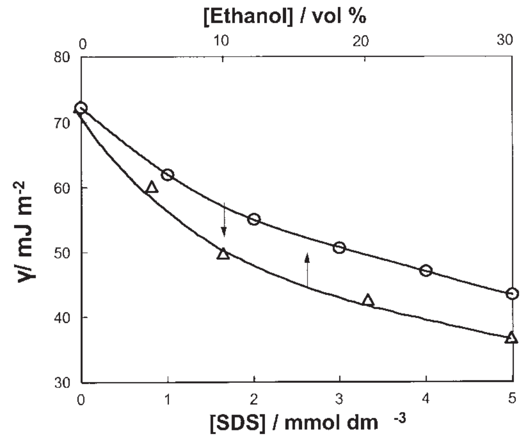

FIG. 5. Surface free energies (γ) of aqueous detergent solutions as functions of SDS (○) and ethanol (△) concentrations. For abbreviation see Figure 1.

strates and arachidic acid are listed in Table 2. These values were calculated from the advancing contact angles of three probe liquids (water, diiodomethane, and ethylene glycol) according to the approach of van Oss, Chaudhury, and Good (18,19). A considerable difference in the base parameter was found among the materials.

Effect of liquid penetration on removal of LB films. Rolling up, emulsification, solubilization, and liquid crystal formation are usual mechanisms in the cleaning of liquid oily contaminants from solid surfaces in aqueous surfactant solutions (3–5). However, rolling up and emulsification do not take place in the oil displacement of solid organic soil such as arachidic acid with high melting point [76.1–76.3°C (20)]. In addition, solubilization and liquid crystal formation do not take place below the CMC in the present system.

The experimental results show that the removal efficiency of arachidic acid from any substrate increased with increasing SDS and ethanol concentrations. In the presence of sodium hydroxide, the removal efficiency further increased. The experimentally determined contact angle and surface free energy decreased with respect to SDS and ethanol concentrations but did not change by the addition of sodium hydroxide. This indicates that the saponification of arachidic acid under alkali conditions (pH 11.2) is a major mechanism of fatty acid removal. However, the dependence of the kind of substrate and the concentrations of SDS and ethanol on the fatty acid removal cannot be explained only by saponification.

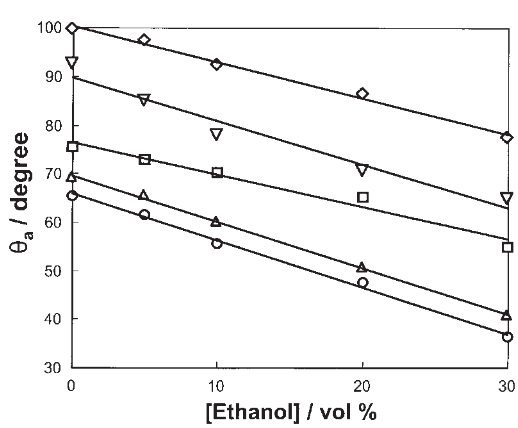

FIG. 4. Advancing contact angles (θ_a) of aqueous detergent solutions on gold (○), polyethylene (◇), nylon 6 (□), cellulose acetate (△), and arachidic acid (▽) as a function of ethanol concentration.

In comparing Figures 1 and 2 with Figures 3, 4, and 5, it is clear that the removal efficiency increased with decreasing contact angle and surface free energy of the detergent solution. This experimental fact suggests that the wetting process by the detergent solution dominates the detachment of fatty acid from the substrate. A thin layer, such as LB films, can be detached directly from the substrate, not removed in gradual steps like ordinary oily contaminants. In this case, the detachment of the LB films from the substrate is equivalent to the penetration of the detergent solution between the LB films and the substrate in the contact zone. Therefore, the results will be discussed in terms of the free energy change due to liquid penetration between the substrate and the LB films.

The free energy change, ΔG, can be calculated as the displacement of the fatty acid/substrate interface by the fatty

TABLE 2
The Lifshitz–van der Waals Component, γ^{LW}, the Acid Parameter, γ^+, and the Base Parameter, γ^-, of Arachidic Acid and the Substrates

Materials	γ^{LW} (mJ m^{-2})	γ^+ (mJ m^{-2})	γ^- (mJ m^{-2})
Arachidic acid	19.6	0.90	2.0
Gold	39.6	0.70	13.8
Cellulose acetate	40.6	0.32	12.4
Nylon 6	38.6	0.60	7.6
Polyethylene	28.6	0.10	0.6

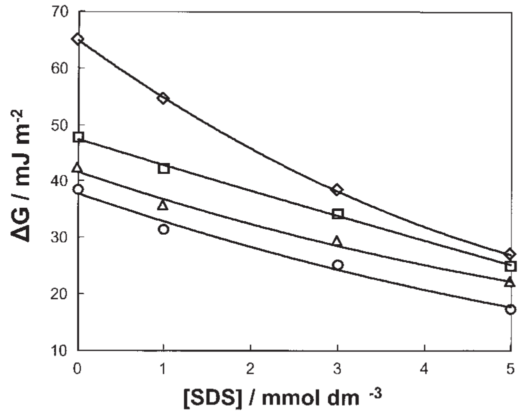

FIG. 6. The calculated free energy change, ΔG, owing to the penetration of aqueous detergent solutions between the arachidic acid Langmuir–Blodgett (LB) films and gold (○), polyethylene (◇), nylon 6 (□), and cellulose acetate (△) substrates as a function of SDS concentration. For abbreviation see Figure 1.

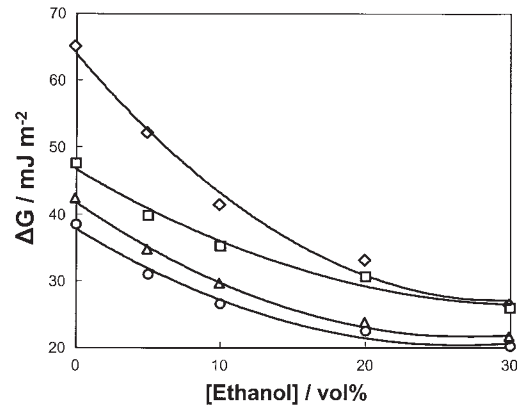

FIG. 7. The calculated free energy change, ΔG, owing to the penetration of aqueous detergent solutions between the arachidic acid LB films and gold (○), polyethylene (◇), nylon 6 (□), or cellulose acetate (△) substrate as a function of ethanol concentration. For abbreviations see Figures 1 and 6.

acid/liquid and the substrate/liquid interfaces (21) as follows (7):

$$\Delta G = \gamma_{AL} + \gamma_{SL} - \gamma_{AS} \quad [1]$$

In this equation, γ_{AL}, γ_{SL}, and γ_{AS} are the interfacial free energies between the arachidic acid films and the detergent solution, the substrate and the detergent solution, and the arachidic acid films and the substrate, respectively. γ_{AL} and γ_{SL} are correlated with the contact angles by the Young equation:

$$\gamma_{AL} = \gamma_A - \gamma_L \cos\theta_{L/A} \quad [2]$$

and

$$\gamma_{SL} = \gamma_S - \gamma_L \cos\theta_{L/S} \quad [3]$$

where γ_A, γ_S, and γ_L are the surface free energies of the arachidic acid films, the substrate, and the detergent solution, respectively. $\theta_{L/A}$ and $\theta_{L/S}$ are the contact angles of the detergent solution on the arachidic acid films and the substrate, respectively. γ_{AS} is rewritten using the surface free energy components as follows (22):

$$\begin{aligned}\gamma_{AS} = {} & \gamma_A^{LW} + \gamma_S^{LW} - 2(\gamma_A^{LW}\gamma_S^{LW})^{1/2} \\ & + 2(\gamma_A^{+}\gamma_A^{-})^{1/2} + 2(\gamma_S^{+}\gamma_S^{-})^{1/2} \\ & - 2(\gamma_A^{+}\gamma_S^{-})^{1/2} - 2(\gamma_S^{+}\gamma_A^{-})^{1/2} \end{aligned} \quad [4]$$

where the superscripts LW, +, and − refer to the Lifshitz–van der Waals component, the Lewis acid parameter, and the Lewis base parameter of the surface free energy, respectively. The value of ΔG was calculated by Equations 1–4 using the advancing contact angles in Figures 3 and 4, the surface free energies of the detergent solutions in Figure 5, and the surface free energy components of arachidic acid and the substrates in Table 2.

The calculated ΔG values are given in Figures 6 and 7 as functions of SDS and ethanol concentrations, respectively. In all cases, the values of ΔG were large and positive, suggesting that fatty acid cannot voluntarily be removed. However, in the present study, fatty acid removal can take place because ultrasonic waves act as the driving force of the removal. ΔG was found to decrease considerably with increasing concentrations of SDS and ethanol. ΔG was smallest for gold, followed by cellulose acetate, nylon 6, and polyethylene. In comparing ΔG in Figures 6 and 7 with the removal efficiency in Figures 1 and 2, the reduction in free energy barrier is expected to be a factor in promoting removal.

Figure 8 demonstrates the relationship between the removal efficiency and ΔG. The results obtained with the three polymer substrates and those with the gold substrate were plotted on the different lines in both the presence (closed symbols) and the absence (open symbols) of sodium hydroxide.

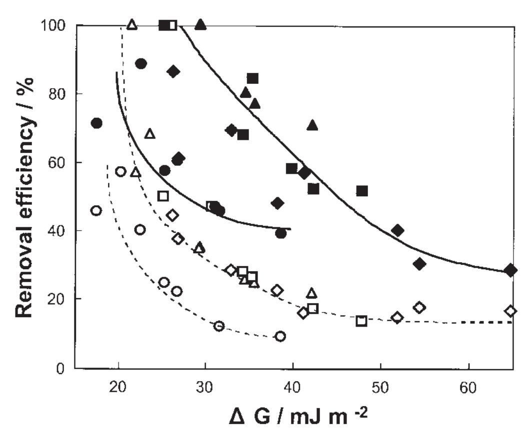

FIG. 8. The relation between the removal efficiency of arachidic acid LB films from gold (○, ●), polyethylene (◇, ◆), nylon 6 (□, ■), and cellulose acetate (△, ▲) substrates in the absence (open symbols) and presence (closed symbols) of 1×10^{-3} mol/dm^3 sodium hydroxide and the free energy change, ΔG, owing to the penetration of aqueous detergent solutions. For abbreviation see Figure 6.

The curves for the gold substrate shifted lower than those for the three polymer substrates. This suggests that additional adhesive force other than the Lifshitz–van der Waals and acid–base interactions operates between arachidic acid and gold. In the absence of sodium hydroxide, a critical value (~25 mJ/m2) of ΔG is required before removal is enhancement. This suggests a threshold barrier below which an opposing driving force due to ultrasonic waves dominates. It is clear that the removal of the LB films of arachidic acid is enhanced by saponification under alkali condition. In all cases, the removal efficiency is found to increase with decreasing ΔG.

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REFERENCES

1. Mamiya, H., Organic Solvent Cleaning and Prevention of Environmental Contamination, *Zairyo-to-Kankyo 40*:122 (1991).
2. Manahan, S.E., *Environmental Chemistry*, 6th edn., CRC Press Inc., Boca Raton, FL, 1994, p. 766.
3. Kabin, J.A., S.L. Tolstedt, A.E. Saez, C.S. Grant, and R.G. Carbonell, Removal of Organic Films from Rotating Disk Using Aqueous Solutions of Nonionic Surfactant: Effect of Surfactant Molecular Structure, *J. Colloid Interface Sci. 206*:102 (1998).
4. Miller, C.A., and K.H. Raney, Solubilization–Emulsification Mechanisms of Detergency, *Colloids Surf. A 74*:169 (1993).
5. Thompson, L., The Role of Oil Detachment Mechanisms in Determining Optimum Detergency Conditions, *J. Colloid Interface Sci. 163*:61 (1994).
6. Lu, C.S., and O. Lewis, Investigation of Film-Thickness Determination by Oscillating Quartz Resonators with Large Mass Load, *J. Appl. Phys. 43*:4385 (1972).
7. Gotoh, K., and M. Tagawa, Detachment Behavior of Langmuir–Blodgett Films of Arachidic Acid from a Gold Surface Studied by the Quartz Crystal Microbalance Method, *Colloids Surf. A 196*:145 (2003).
8. Gotoh, K., and Y. Nakata, Detergency of Oily Contaminant as Assessed with a Quartz Crystal Microbalance and Langmuir–Blodgett Methods, *J. Oleo Sci. 53*:489 (2004).
9. Hann, R.A., Molecular Structure and Monolayer Properties, in *Langmuir–Blodgett Films*, edited by G. Robert, Plenum Press, New York, 1990, p. 17.
10. Ariga, K., and Y. Okahata, *In situ* Characterization of Langmuir–Blodgett Films During a Transfer Process. Evaluation of Transfer Ratio and Water Incorporation by Using a Quartz Crystal Microbalance, *Langmuir 10*:3255 (1994).
11. Sauerbrey, G., Verwendung von Schwingquarzen zur Wagung dunner Schichten und zur Mikrowagung, *Z. Physik. 155*:206 (1959).
12. Gotoh, K., N. Nakazawa, and M. Tagawa, Applying the Langmuir–Blodgett Technique to a Well Defined Model Detergent System, *Textile Res. J. 71*:481 (2001).
13. Shimomura, K., H. Onozawa, and J. Komiyama, Protein Soil Release Using Protease as Monitored with a Quartz Crystal Microbalance, *Textile Res. J. 67*:348 (1997).
14. Weerawardena, A., C.J. Drummond, F. Caruso, and M. McCormick, Real Time Monitoring of the Detergency Process by Using Quartz Crystal Microbalance, *Langmuir 14*:575 (1998).
15. Daikhin, L., and M. Urbakh, Effect of Surface Film Structure on the Quartz Crystal Microbalance Response in Liquids, *Langmuir 12*:6354 (1996).
16. Gotoh, K., M. Tagawa, N. Ohmae, H. Kinoshita, and M. Tagawa, Surface Characterization of Atomic Oxygen Beam-Exposed Polyimide Films Using Contact Angle Measurements, *Colloid Polym. Sci. 279*:214 (2001).
17. MacLead, C.A., and C.J. Radke, A Growing Drop Technique for Measuring Dynamic Interfacial Tension, *J. Colloid Interface Sci. 160*:435 (1993).
18. Gotoh, K., J. Tao, and M. Tagawa, Adhesion Interaction in Water/*n*-Alcohol Mixtures Between Silanized Silica and Polymer Particles, *J. Adhesion Sci. Technol. 13*:1307 (1999).
19. van Oss, C.J., R.J. Good, and M.K. Chaudhury, Additive and Nonadditive Surface Tension Components and the Interpretation of Contact Angles, *Langmuir 4*:884 (1988).
20. Ishii, S., *et al.*, in *The Chemical Handbook I*, edited by M. Ohki *et al.*, Maruzen, Tokyo, 1993, p. 254.
21. Lange, H., Physical Chemistry of Cleaning Action, in *Solvent Properties of Surfactant Solutions*, edited by K. Shinoda, Marcel Dekker, New York, 1967, p. 156.
22. van Oss, C.J., R.F. Giese, and P.M. Costanzo, DLVO and Non-DLVO Interactions in Hectorite, *Clay Clay Miner. 38*:151 (1990).

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